

Real Analysis I — Homework 4

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Unless otherwise specified, all functions are defined on $E \subset \mathbb{R}^n$ where $E \in \mathcal{M}$.

For notation simplicity, a set such as $\{x \in E : f(x) > t\}$ is written as $\{f > t\}$ for short.

1. Suppose $f_k \xrightarrow{\mu} f$ on E , and $f_k \leq g$ a.e. E . Show that $f \leq g$ a.e. E .
2. Suppose $f_k \xrightarrow{\mu} f$ on E , and $f_k \leq f_{k+1}$ a.e. E for all $k \in \mathbb{N}$. Show that $f_k \rightarrow f$ a.e. E .
3. Find the integral $\int_0^1 f$ where

$$f(x) = \begin{cases} \frac{1}{\sqrt{x}} & \text{if } x \notin \mathbb{Q} \\ x^3 & \text{if } x \in \mathbb{Q} \end{cases}$$

4. Suppose $f, g \in L(E)$. Prove that $\sqrt{|f|^2 + |g|^2} \in L(E)$.
5. Suppose $f \in L(E)$, and for any bounded measurable function $\phi : E \rightarrow \mathbb{R}$ there is $\int_E f\phi = 0$. Show that $f = 0$ a.e. E .
6. Suppose $f \in L(E)$ and $\mu(E) < \infty$. Show that for any $\epsilon > 0$, there exists a bounded measurable function $g : E \rightarrow \mathbb{R}$ such that $\int_E |f - g| < \epsilon$.
7. Suppose $\mu(E) < \infty$. Show that $f \in L(E)$ iff $\sum_{k=0}^{\infty} \mu(\{|f| \geq k\}) < \infty$.
8. Prove that for any $p > -1$ there is

$$\int_0^1 \frac{x^p}{1-x} \ln \frac{1}{x} dx = \sum_{k=1}^{\infty} \frac{1}{(p+k)^2}$$

9. Prove that

$$\lim_{k \rightarrow \infty} \int_0^{\infty} \frac{1}{(1 + \frac{t}{k})^k \cdot t^{1/k}} dt = 1$$

10. Suppose f is integrable on $(0, \infty)$. Define

$$g(x) = \int_0^{\infty} \frac{f(t)}{x+t} dt$$

for any $x \in (0, \infty)$. Is g continuous on $(0, \infty)$? Justify your answer.